A replication of Chumney (2013): four estimators for structural equation models with small samples

29 October 2026

Abstract

Which estimator recovers a three-factor structural equation model when the sample is small? Chumney (2013) compared maximum likelihood, partial least squares, generalized structured component analysis, and Bayesian Markov chain Monte Carlo on normal data, and concluded that maximum likelihood is generally the most robust of the four and the approach of choice for applied researchers. We regenerated the 36 data conditions in the dissertation and refit maximum likelihood, partial least squares, and generalized structured component analysis, with 150 replications in each cell. The ordering of percent bias in the measurement and structural parameters matches the printed tables for maximum likelihood and partial least squares, including the cells in which partial least squares is less biased. The R implementation of generalized structured component analysis matches those tables at three and five indicators per factor and misses the printed seven-indicator column. The Bayesian arm was not refit.

1 The headline

Does maximum likelihood dominate the other three estimators on normal data from this three-factor model? In the slice we refit, it dominates partial least squares and generalized structured component analysis on measurement percent bias when the indicators are reflective, and the dissertation's own exceptions are reproduced. The Bayesian estimator was not refit, so the headline sentence, which names four methods, is a partial match.

Chumney's dissertation asks how sample size, the number of items, omitted cross-loadings, and reflective versus formative indicators change the behavior of four estimators. The closing paragraph of Chapter 5 states the result that the abstract also states. Figure 1 is that paragraph. The sentence is: maximum likelihood is generally the most robust of the four, and it should be the approach of choice, with the dependence on the normal data and the models in the study kept in view.

Table 1 is the comparison. A match means the difference of Monte Carlo means lies inside a two-standard-error band and is not rejected by a Holm adjustment within that outcome on the three-indicator slice. Percent bias is equation (15), the absolute relative error times 100, averaged over the loadings or over the two paths. The goodness-of-fit row uses the benchmark $\sqrt{1 \times 0.36} = 0.600$, discussed in Section 4. Counts of converged replications are out of 150.

The first measurement row is the headline comparison in numbers. At $n = 1000$ with three reflective indicators and a correct model, maximum likelihood's mean absolute percent bias is about 2.7 and the two component methods sit at about 15.9. Our means are 2.730, 15.919, and 15.938. Section 3 records the same ordering at five and seven indicators for maximum likelihood and partial least squares.

> analyzed and interpreted for the purposes of evaluating a study or developing subsequent projects. The results of the present study indicate that ML is generally the most robust of the four estimation methods studied, and should therefore be the approach of choice for applied researchers. It is important to note, however, that such a conclusion is dependent upon the characteristics of the data and models used for the present study, and this recommendation may not generalize to other contexts.

Figure 1: Chumney (2013), dissertation page 118. The headline sentence is the one that names maximum likelihood as generally the most robust of the four methods.

The misspecified reflective structural rows are the exception Chapter 5 states in words: partial least squares and generalized structured component analysis are less biased than maximum likelihood for the paths when cross-loadings are omitted. Our means keep that order, 7.884 and 7.917 against 15.203, next to the printed 7.823, 7.791, and 15.652.

The goodness-of-fit pair is one printed number, -0.180, under two benchmarks. Subtracting 0.600 from our mean goodness of fit reproduces -0.179. Subtracting the standardized population value 0.420 produces $+0.001$. Section 4 says what 0.600 is.

The formative structural row for maximum likelihood is a real gap in an otherwise matched cell. Our partial least squares mean in that same cell is 2.605, next to the printed partial least squares value 2.768, while the printed maximum likelihood value is 1.196. Our two estimators agree with each other and with the printed partial least squares cell.

The seven-indicator generalized structured component analysis rows are the place the R package and the printed column separate. At three indicators the same package matches the printed measurement bias of 15.918 with 15.938. At seven indicators the printed measurement bias rises to 13.667, and our mean falls to 7.189, in line with our partial least squares mean of 7.184 and the printed partial least squares mean of 7.064.

2 What we did ourselves, and what we took from others

This section answers what is ours and what is Chumney's. The model, the $3 \times 3 \times 2 \times 2 \times 4$ design, and Tables 1–9 are the dissertation's. The population covariance, the 150-replication Monte Carlo, and Table 1 are ours.

The population model is a chain of three factors. The paths are 0.60. In the marker scaling used to draw the data, each factor variance is 0.49, each structural disturbance variance is 0.3136, and a simple reflective loading is 1 with unique variance 0.51. Those numbers are the unstandardized parameters whose standardized values are the ones named in the text: loadings 0.70, unique variances 0.51, disturbances 0.64. Figure 2 is the paragraph that states the standardized values. The figure above that paragraph prints the marker scaling, including a disturbance of 0.31, which

is 0.3136 rounded to two decimals.

> For the conditions of correct model specification with reflective indicators, the population models specify that all indicator-latent variable relationships have standardized values of 0.700, the error terms associated with all indicators have standardized values of 0.510, the error terms associated with the latent variables have standardized values of 1.000 (η_1) and 0.640 (η_2 and η_3), the path coefficients linking the latent variables each have standardized values of 0.600, and the models do not include

Figure 2: Chumney (2013), dissertation page 38. Standardized population values for the correctly specified reflective model.

Formative conditions are reflective indicators with unique variance 0.001. The dissertation says so: the formative arm is an almost error-free reflective model, so that every estimator, including partial least squares in mode A, can be applied in the same way. Misspecification omits three cross-loadings that are in the population. The analysis model is always simple structure. Section **??** records which items those are and which unstandardized cross-loading reproduces the parenthetical 0.21.

Data were drawn from the multivariate normal distribution with that covariance, 150 times in each of the 36 conditions. The dissertation drew the same conditions in Mplus 6. We did not have Mplus 6. Seeds are `simulation/replicate.R:40`. Replication r in a cell uses `set.seed(cell_id + r)`.

Maximum likelihood was fit with lavaan 0.7-3, factor variances fixed at 1, at most 1000 iterations (`simulation/replicate.R:58`). A replication counts as converged when the optimizer converges and every unique variance and factor residual variance is positive. Partial least squares was fit with cSEM 0.7.1, mode A, centroid inner weights, classical outer loadings, at most 1000 iterations (`simulation/replicate.R:94`). Generalized structured component analysis was fit with gesca 1.0.5, at most 1000 iterations and no bootstrap (`simulation/replicate.R:131`). A component loading outside $[-1, 1]$ was treated as an implausible residual, which did not occur. Percent bias is `simulation/replicate.R:173`, the average of equation (15).

The tools were chosen for the following reasons. R 4.3.3 (`https://cran.r-project.org/`) is the language the partial least squares package in the dissertation runs in. lavaan 0.7-3 (`https://cran.r-project.org/package=lavaan`) estimates the same normal maximum likelihood the dissertation estimated in Mplus. cSEM 0.7.1 (`https://cran.r-project.org/package=cSEM`) estimates partial least squares path models. The dissertation cites the plsSEM package of Monecke and Leisch (2012), which is semPLS on CRAN. semPLS 1.0-10 installs, and `plsm` then errors on this R because `class()` of a matrix has length 2 and is combined with `||`. cSEM with disattenuation turned off returns the classical outer correlations that semPLS would have returned. gesca 1.0.5 (`https://cran.r-project.org/package=gesca`) is an R implementation of generalized structured component analysis. The dissertation used the GeSCA program at `http://www.sem-gesca.org`.

That host did not resolve on 29 October 2026. mvtnorm 1.4-2 (https://cran.r-project.org/package=mvtnorm) draws the normal samples. The full grid of 36 conditions and three estimators took 929 seconds and is stored in simulation/out/reps150-full.rds.

The Bayesian arm is outstanding. blavaan 0.6-1 depends on rstan, and that toolchain was not built here. The dissertation estimates the Bayesian model in Mplus 6, names a maximum of 1000 iterations, and includes Markov chain Monte Carlo in the parenthetical list of methods that use 500 bootstrap samples for standard errors. Priors, the number of chains, and whether the standard errors are posterior standard deviations or bootstrap standard errors are not stated beyond that sentence.

3 What matched

Do the printed percent biases reappear under the population values in Figure 2? For maximum likelihood and partial least squares, yes, across the item counts we checked. For generalized structured component analysis, yes at three and five indicators per factor.

At $n = 1000$, correctly specified reflective indicators, maximum likelihood measurement percent bias is 2.730, 2.234, and 2.085 at three, five, and seven items. The printed values are 2.712, 2.261, and 2.117. Partial least squares is 15.919, 9.841, and 7.184, next to 15.896, 9.715, and 7.064. The three-indicator rows are inside the Holm band in simulation/out/comparison_i3.csv. The five- and seven-item maximum likelihood and partial least squares rows move with the printed column by a few tenths, on a scale where the standard deviations in Table 3 are about 0.4 to 0.8 at $n = 1000$.

Partial least squares and generalized structured component analysis each converged on all 5400 replications. The dissertation reports the same 5400/5400. Maximum likelihood converged on 4648 of 5400 replications, which is 86.1 percent. The prose in Chapter 4 says 94 percent. The sum of Table 1 is 5038 of 5400, which is 93.3 percent. Our shortfall relative to Table 1 is concentrated in the formative misspecified cells with five and seven items, and in one reflective cell discussed in Section 7. At three items the misspecified formative counts are 21 against 35 at $n = 50$, and 102 against 119 at $n = 1000$. Those two differences have z of 2.09 and 2.25. Holm within the convergence family labels them close.

On the three-indicator slice, 20 of 24 measurement comparisons and 22 of 24 structural comparisons are matches after Holm. The two measurement rows labeled close in the output are partial least squares and generalized structured component analysis under misspecified reflective indicators at $n = 1000$: 19.377 and 19.371 against a printed 19.196, with z about 2.4. The absolute gaps are under 0.2 on a percent-bias scale whose interesting contrasts are several points wide.

4 What the goodness-of-fit bias required

Which population goodness of fit is being subtracted in Table 2? For correctly specified reflective models, the subtraction that reproduces the printed bias uses 0.600. The subtraction that uses equations (12)–(14) on the standardized loadings does not.

Equation (14) is the square root of average communality times average R^2. Equation (12) defines a block communality as the average squared correlation of the indicators with the component. For the standardized reflective model every correlation is 0.70, so the average communality is 0.49, the endogenous R^2 values are 0.36, and the goodness of fit is $\sqrt{0.49 \times 0.36} = 0.420$. Our maximum likelihood mean at $n = 1000$ with three reflective indicators is 0.421. The printed bias is -0.180. The difference $0.421 - 0.600 = -0.179$ falls inside the band around -0.180. The value 0.600 is

$\sqrt{1 \times 0.36}$: communality equal to the square of the unstandardized loading 1 that the figure prints above the parentheses, times the endogenous R^2 of 0.36.

The same 0.600 benchmark puts our partial least squares bias at -0.237 next to the printed -0.238. Our gesca bias in that cell is -0.237 next to the printed -0.251. The absolute gap is 0.014. The printed standard deviation in that gesca cell is 0.002, so the two-standard-error band is narrow and the verdict is no match. Our standard deviation is 0.016, in line with partial least squares in the same cell. The printed generalized structured component analysis column is less variable than either component method was in our run.

Misspecified cells do not fall into place under either benchmark. At $n = 1000$, three reflective indicators, omitted cross-loadings, our maximum likelihood goodness of fit is 0.516. Minus 0.600 that is -0.084. Minus the standardized population value 0.447 it is $+0.070$. The printed bias is $+0.147$. Both readings miss, and the same sign error appears for partial least squares and gesca. Section 7 keeps that gap open.

Formative models have a standardized loading near 1, so 0.600 and the standardized population value 0.599 agree. Printed biases in the correctly specified formative cells are within a few thousandths of zero, and our biases are as well.

5 Assumptions

Each subsection is a choice the dissertation does not pin down and that we supplied. Turning the choice off is the check named in the subsection.

5.1 Disturbance variance 0.3136

The figure prints a structural disturbance of 0.31 under a standardized value of 0.64. With factor variance 0.49, the disturbance that keeps both factor variances equal and the standardized disturbance equal to 0.64 is $0.49 \times 0.64 = 0.3136$. We used 0.3136. Using the printed 0.31 would make the second factor variance 0.4864 instead of 0.49. The standardized loading would move by less than 0.01. The match of Table 3 at $n = 1000$ to two or three decimals is the check that the rounding does not carry the comparison.

5.2 Nominal targets in equation (15)

Equation (15) divides by the population parameter. We used the standardized values named in the text: 0.70 for every reflective loading, 1.00 for a formative loading on a single-factor item, 0.56 for a formative loading on a cross-loading item, and 0.60 for each path. The completely standardized formative loading implied by unique variance 0.001 is 0.999. Comparing estimates with 1.00 leaves a percent bias of about 0.10 that does not shrink with n. Table 3 prints 0.100 for maximum likelihood in the correctly specified formative cell at $n = 1000$, with a standard deviation printed as 0.000. Our mean is 0.102. A z-test was not computed, because the printed standard deviation does not give a positive standard error. The gap of 0.002 is the difference between 0.102 and a three-decimal printing of a quantity that the marker model puts at 0.102.

5.3 Centroid weights, classical outer loadings

semPLS defaults to the centroid scheme. cSEM's centroid scheme was the one we ran. On one sample of 1000 from the reflective model, centroid and path weights returned the same loadings and the same paths, which is what a chain with a single predictor for each endogenous factor leads

one to expect. Disattenuation was left off. Turning it on pulled the outer loadings from about 0.81 back toward 0.70 and destroyed the match to the printed partial least squares percent bias of 15.9.

5.4 Which replications are eligible for percent bias

Percent bias is averaged over replications that converged to positive residual variances. The dissertation uses the same rule. Where our convergence count is lower, the bias is the bias among the replications lavaan accepted. At three formative misspecified indicators and $n = 1000$, that subset is 102 replications against the printed 119, and the measurement percent bias still matches (21.686 against 21.643).

6 Ambiguities that were resolved

6.1 The cross-loading labeled 1 (0.21)

Figure 3 is the arrow from η_1 to Y8 in Figure 3 of the dissertation. The label on the arrow is 1 above and (0.21) below. The primary loading on Y8 is 1 above and (0.70) below, and the unique variance is 0.29. Those three parenthetical and residual values are mutually consistent when the unstandardized cross-loading is 0.30, the factor variance is 0.49, and the primary unstandardized loading is 1. An unstandardized cross-loading of 1, the number printed above (0.21), does not produce a standardized cross-loading of 0.21 together with a standardized primary loading of 0.70.

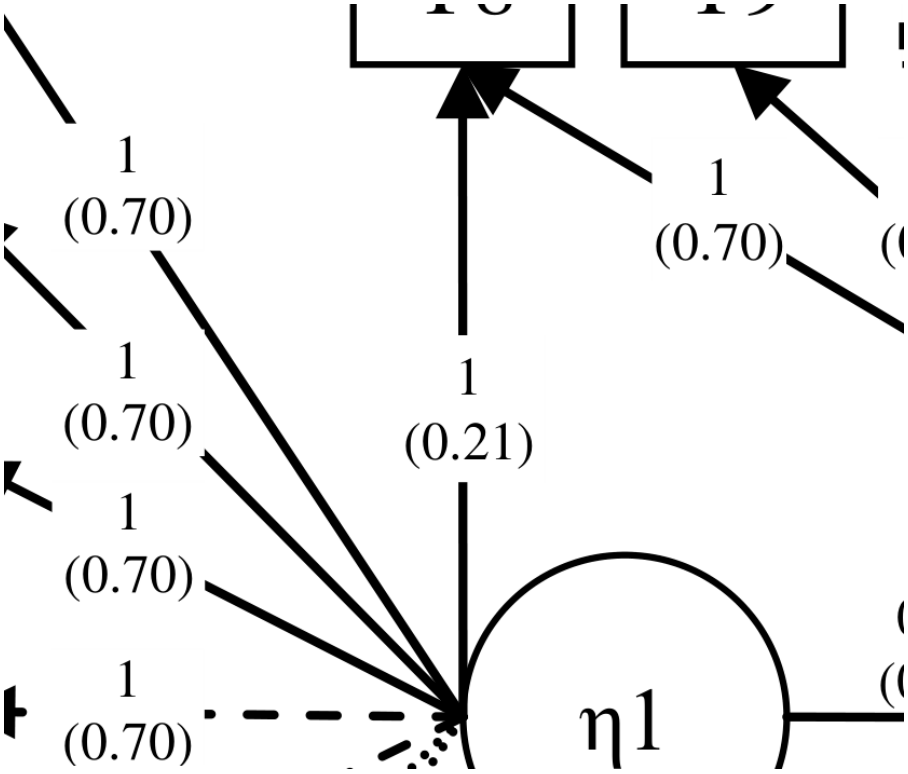


Figure 3: Chumney (2013), Figure 3, the cross-loading of Y8 on η_1. The crop is 600 dpi, from `paper/figures/crop_pages.py`.

We drew the data with the unstandardized cross-loading 0.30. The standardized values that fall out are 0.700 and 0.210, up to rounding from a unique variance of 0.29 (`simulation/population.R`,

simulation/population.R:47; simulation/check_standardized.R). The command is `Rscript simulation/check_standardized.R`, which writes `simulation/out/standardized_check.json`. The check against the dissertation is the misspecified reflective measurement row of Table 1: printed 9.673, ours 9.676.

For formative misspecification the figure prints an unstandardized 1 on both arrows of a cross-loading item, and both parenthetical standardized values are 0.56. That pair is consistent. Our measurement percent bias in that cell is 21.686 against the printed 21.643.

6.2 Which indicators belong to the three-item model

The note under Figures 1–4 says solid lines are in every condition, dashed lines are added at five items, and dotted lines are added at seven. The solid boxes on η_2 are Y8, Y9, and Y14: the first indicator, the second, and the last. Y8, Y14, and Y15 are the three cross-loading items, and they are solid, so the cross-loadings are present at every item count. The indicator sets are simulation/population.R:6. The 600 dpi crop of the population paragraph is `paper/figures/population.png`, regenerated by `python3 paper/figures/crop_pages.py`.

6.3 Equation (15) is a percent

The text layer of the PDF drops the fraction bar and the absolute-value bars in equation (15). The page shows $\%\text{BIAS} = [|\theta_i - \theta_B|/\theta_B] \times 100$. Figure 4 is the crop. Reading the formula as a proportion rather than a percent would miss Table 3 by a factor of 100. The crop is `paper/figures/equation15.png`.

(Enders & Bandalos, 2001), and is calculated

$$\%BIAS = \left[\frac{|\theta_i - \theta_B|}{\theta_B}\right] \times 100 \qquad (15)$$

where θ_i is the recovered parameter estimate and θ_B is the known population parameter.

Figure 4: Chumney (2013), equation (15), dissertation page 47 in the PDF numbering of the body. Absolute percent bias.

7 What remains unresolved

The Bayesian comparison is open. The dissertation's Markov chain Monte Carlo convergence rate in the prose is 87.7 percent, and the sum of Table 1 is 4769 of 5400, which is 88.3 percent. We did not refit that arm. A blavaan run with a stated prior, a stated iteration cap of 1000, and the same seeds would settle whether the posterior means track maximum likelihood as closely as Table 3's pairing of the two methods suggests.

The goodness-of-fit bias under misspecification is open. Both benchmarks in Section 4 miss the printed positive biases, by about 0.08 to 0.23 depending on the cell and the benchmark. The parameter estimates in those same cells match, so the gap is in the goodness-of-fit number rather than in the loadings. What would settle it is the program that computed the "true" goodness of fit, or a record of whether η_1 was given an R^2 of 0 inside that average. The prose says the average runs

over η_1, η_2, and η_3. Putting a zero in for η_1 produces $\sqrt{\bar{C} \times 0.24}$. For the misspecified reflective population, $\bar{C} = 0.554$ and that square root is 0.365. Our estimated goodness of fit of 0.516 minus 0.365 is 0.151, next to the printed 0.147. That reading fits this one cell and does not fit the correctly specified reflective cell, where the same rule would predict a bias near +0.08 and the table prints −0.180. One formula has not covered both cells.

Maximum likelihood convergence under formative misspecification with five and seven items is lower in our run than in Table 1, by 30 to 100 replications in a cell. At $n = 50$ and seven items the table shows 134 of 150 and we accepted 34. The unique variance in the population is 0.001 and each cross-loading item has two unstandardized loadings of 1, so lavaan's nonpositive-variance check fires often. Mplus 6's check on the same covariance may treat the boundary differently. The measurement percent bias among the replications we did accept still tracks Table 3 at three items. The original Mplus input, or the saved improper-solution flags, would settle the count.

One reflective cell goes the other way. At $n = 50$, seven items, omitted cross-loadings, Table 1 shows 102 converged maximum likelihood replications and we accepted 150. The z against a binomial standard error is about 8. The same population values reproduce the measurement percent bias at $n = 1000$ in that specification, so the covariance is not wildly wrong. The open piece is again the improper-solution rule at the smallest sample.

The seven-indicator generalized structured component analysis column remains open as a software difference. gesca 1.0.5 tracks the printed column at three and five indicators and then keeps declining, as partial least squares does. The printed column rises at seven indicators: measurement percent bias 13.667 against our 7.189, structural percent bias 4.747 against our 12.811. Our structural mean of 12.811 sits next to our partial least squares mean of 12.747 and the printed partial least squares mean of 12.153. A run of the 2013 GeSCA program on the same draws would show whether the printed uptick is the web program. The host named in the dissertation did not resolve.

The printed structural percent bias 4.747 in that seven-indicator cell is also the printed structural percent bias for generalized structured component analysis at $n = 300$, seven formative indicators, in the same table. Whether one cell was typeset twice is not settled by the page. The cell is a no match under our reading either way.

8 A short list of internal checks

These are comparisons inside the dissertation, kept separate from Table 1. They are not, by themselves, failures of the Monte Carlo.

The prose says maximum likelihood converged for 94 percent of replications and Markov chain Monte Carlo for 87.7 percent. Adding Table 1 gives 5038 of 5400 (93.3 percent) and 4769 of 5400 (88.3 percent). The differences are 38 and 34 replications. A misread digit in one cell would close either gap. The 600 dpi crop of the table is `paper/figures/table1.png`.

The printed standard deviation of the formative maximum likelihood measurement bias at $n = 1000$ is 0.000. Our standard deviation is 0.003. The means are 0.100 and 0.102.

9 Code and source availability

The dissertation names Mplus 6, the plsSEM package, and GeSCA at `http://www.sem-gesca.org`. It does not name a repository for the simulation script. The checks below were made on 29 October 2026.

Place checked	UTC date	Result
Dissertation PDF, procedures and references	2026-10-29	Obtained. The PDF contains no program file.
Digital Commons record cehsdiss/189	2026-10-29	HTML 200. The PDF is the file. No separate code file.
Wayback raw file of the viewcontent PDF	2026-10-29	PDF obtained, 146 pages.
sem-gesca.org	2026-10-29	The host name did not resolve.
GitHub user search, Frances Chumney	2026-10-29	No user. No repository to clone.
GitHub repository search for chumney	2026-10-29	Unrelated repositories only.
GitHub code search for the dissertation	2026-10-29	HTTP 429. The search did not finish.
OSF search for the title and for the author	2026-10-29	No records and no user.
Zenodo title search	2026-10-29	No records.
GitLab project search	2026-10-29	No projects.
CRAN semPLS 1.0-10	2026-10-29	Source installs. `plsm` errors on R 4.3.

The PDF used for the crops is `local-extraction/literature/unl-sem/chumney2013.pdf`. The file stays outside git. The sha256 is `1b067efef410ac3ae412e3a26a7b29bf1f8e50e935d219353fbe16ce8a853007`.

10 Next steps

The missing piece for the headline's fourth method is a Bayesian refit with a stated prior and an iteration cap of 1000, on the same seeds, scored with equation (15). blavaan is the R package set up for that comparison. The missing piece for the seven-indicator generalized structured component analysis column is a run in the GeSCA program the dissertation used, on draws from `simulation/population.R`. The missing piece for the goodness-of-fit sign under misspecification is the code that computed the population goodness of fit, or the Mplus output that stored it.

Appendix: artifacts

Claim	File	Command
Table 1, three-indicator bands	`simulation/out/comparison_i3.csv`	`Rscriptsimulation/compare.R`
All 36 cells, three estimators	`simulation/out/reps150-full.rds`	`Rscriptsimulation/replicate.R--reps150`
Standardized loadings 0.70 and 0.21	`simulation/out/standardized_check.json`	`Rscriptsimulation/check_standardized.R`
Headline, Table 1, equation (15), cross-loading	`paper/figures/*.png`	`python3paper/figures/crop_pages.py`

Commands are documented in `README.md`. `R_LIBS_USER` must contain lavaan, cSEM, gesca, and mvtnorm. The dissertation PDF is an argument to `crop_pages.py` and is not in git.

References

[1] Frances L. Chumney. Structural equation models with small samples: A comparative study of four approaches. Ph.D. dissertation, University of Nebraska–Lincoln, 2013. https://digitalcommons.unl.edu/cehsdiss/189.

[2] Armin Monecke and Friedrich Leisch. semPLS: Structural equation modeling using partial least squares. *Journal of Statistical Software*, 48(3):1–32, 2012.

Table 1: Printed claims and our Monte Carlo means. The three-indicator cells are from Tables 1–4. The seven-indicator cells are from Tables 3 and 4. Standard errors and the Holm labels are in simulation/out/comparison_i3.csv.

Claim	Published	Ours	Verdict
ML measurement bias, $n = 1000$, correct, reflective	2.712	2.730	Match
PLS measurement bias, same cell	15.896	15.919	Match
GSCA measurement bias, same cell	15.918	15.938	Match
ML structural bias, same cell	4.203	4.106	Match
PLS structural bias, misspecified reflective, $n = 1000$	7.823	7.884	Match
ML structural bias, same misspecified cell	15.652	15.203	Match
ML measurement bias, misspecified reflective, $n = 1000$	9.673	9.676	Match
ML measurement bias, misspecified formative, $n = 1000$	21.643	21.686	Match
PLS and GSCA convergence, all 36 conditions	5400/5400	5400/5400	Match
ML GoF bias, correct reflective, $n = 1000$, minus 0.600	-0.180	-0.179	Match
ML GoF bias, same cell, minus the standardized population value 0.420	-0.180	$+0.001$	No match
ML convergence, misspecified formative, $n = 50$	35	21	Close
ML convergence, misspecified formative, $n = 1000$	119	102	Close
ML structural bias, correct formative, $n = 1000$	1.196	2.601	No match
GSCA measurement bias, correct reflective, $n = 1000$, seven items	13.667	7.189	No match
GSCA structural bias, same seven-item cell	4.747	12.811	No match
ML convergence, misspecified reflective, $n = 50$, seven items	102	150	No match